

Seminar 4

1. Introduction

The goal of the exercise is to analyse the relationship between the mean population and mean GDP per capita in the period 2001-2019.

The main question is whether there is a linear relationship between a country's demographic dimension and the level of economic well-being.

The two datasets used were the Global Population dataset and the Global GDP dataset.

The two final variables used during the analysis were mean population and mean GDP per capita .

2. Data pre-processing

To prepare the data, the following process was done:

- Selected columns corresponding to the years 2001 to 2019
- Converted the selected columns to numeric values
- Handled the NaN through interpolation and forward/backward fill
- Deleted rows with insufficient data
- Calculated the mean for each country

The final dataset contained the mean population and the mean GDP per capita for each country.

3. Exploratory Analysis and Correlation

The Pearson correlation between the population means and GDP per capita means resulted in -0.03. As the correlation coefficient is extremely close to 0, this indicates the absence of a significant linear relationship between a country's population and its GDP per capita.

4. Linear Regression

The linear regression model was applied with the mean population (X), which is an independent variable, and mean GDP per capita (Y), which is a dependent variable, and the dataset was divided into a training and test set.

The results were:

- Coefficient – 0.0036
- Intercept: 10.18
- Mean Absolute Error (MAE): 15,986,035.24
- R2: - 0.001

The negative R2 value means that the model is not able to explain the GDP per capita variability. This suggests that there is no linear relationship between the analysed variables.

5. Conclusion

The results show that population is not a determining factor for the GDP per capita, probably because a country's economic development also depends on structural factors and the demographic dimension is not sufficient to explain all the differences.

Based on the statistical analysis, it is possible to state that the correlation between population and GDP per capita is close to 0 (-0.03) and the linear regression model has no predictive capacity (R2 -0.001).